

PixelRoot: Hardware-Anchored, Sensor-Level Media Provenance with Real-Time Blockchain Notarization for Trustworthy Imaging

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Abstract—The realism of AI-generated “deepfake” imagery has outpaced society’s ability to tell synthetic media from camera-captured reality. Detector-based defenses are locked in an adversarial arms race and generalize poorly to unseen generators, while existing provenance schemes bind trust to meta-data that can be stripped, edited, or transplanted. We present PixelRoot, a framework that moves the root of trust from meta-data into the pixels themselves. At capture, a CMOS image sensor applies a keyed, spread-spectrum modulation—realized by a small, low-frequency per-column/per-region programmable-gain profile that today’s column-parallel sensors can already approximate—to embed a 128-bit, error-corrected provenance payload (device identity, trusted timestamp, optional location). A SHA-256 commitment over the frame is simultaneously notarized to a public blockchain within seconds of the shutter, closing the window for substitution. We contribute: (i) a structured survey unifying deepfake detection, PRNU forensics, watermarking, C2PA, and blockchain media integrity into a dual-binding design; (ii) a formal architecture, threat model, and security argument with three propositions; (iii) a measured robustness study on real public images (Kodak) with real libjpeg/WebP codecs and a real Reed–Solomon code under a WAVES-style attack suite—recovering the payload on 23/24 images down to JPEG Q20 (21/24 at Q10) at 38 dB PSNR, with 0/1224 false-accepts and a documented geometric limitation; (iv) a CMOS-manufacturing feasibility analysis of per-pixel gain control, with engineering schematics, and the tamper-localization layer it unlocks (ROC-AUC 0.97); and (v) a privacy-preserving notarization design and a phased adoption path. Deployed as an OEM interoperability standard akin to JPEG, sensor-level provenance plus real-time notarization can give legal, journalistic, and safety-critical applications an affirmative, hardware-rooted proof of authenticity rather than a probabilistic guess.

Index Terms—media provenance, deepfakes, blockchain, CMOS image sensor, PRNU, digital watermarking, content authenticity, C2PA, image forensics, JPEG.

I. Introduction

DIGITAL images and videos have become primary evidence in journalism, courts, insurance, humanitarian monitoring, and everyday social discourse. That evidentiary role is collapsing. Diffusion models, generative adversarial networks, and neural rendering now synthesize photorealistic faces, scenes, and full videos that humans—and automated detectors—routinely fail to distinguish from reality [1], [2]. The harm is no longer hypothetical.

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Industry incident trackers attribute on the order of \$1.2–1.3 billion in documented fraud losses to deepfakes in 2025, roughly tripling year over year, with the overwhelming majority of losses originating on social-media platforms [3], [4]; enterprise surveys report that a majority of organizations experienced deepfake-related incidents with mean losses exceeding \$280,000 [5]. Beyond fraud, non-consensual intimate imagery and election disinformation erode personal safety and institutional trust.

The dominant technical response has been reactive detection: train a classifier to spot the statistical fingerprints of a given generator. This strategy is structurally disadvantaged. Detectors overfit to the artifacts of the generators in their training set and degrade sharply on unseen models—recent in-the-wild benchmarks report AUC drops of roughly 45–50% relative to academic datasets [1], [6]. Because every published detector also serves as a differentiable signal that the next generation of generators can be trained to defeat, detection is a perpetual “tug of war” that the defender is not guaranteed to win [7], [8].

An alternative philosophy inverts the problem: instead of trying to prove a piece of media is fake, prove that a piece of media is real by binding it, at creation time, to a trustworthy origin. This is the provenance paradigm now embodied by the C2PA Content Credentials standard and by secure-capture products such as Truepic and ProofMode [9]–[12]. These systems are a major advance—C2PA supports hardware-backed signing, rich edit-history assertions, and ingredient hashes, and our goal is to complement rather than replace it. But the authenticity claim is carried, in the common case, in a signed metadata container attached to the file, and that container is a single point of failure: metadata can be deliberately stripped by platforms, lost in transcoding, or, worse, the binding itself can be attacked. In September 2025, a vulnerability in the Nikon Z6 III’s C2PA implementation let a camera sign fabricated composites as authentic, forcing the vendor to suspend its authenticity service and revoke every certificate it had issued [13]. If the credential is removed, the pixels carry no inherent evidence of their origin. PixelRoot targets exactly this gap, and can populate C2PA assertions while adding an in-pixel, ledger-witnessed binding that survives container loss.

A. The PixelRoot thesis

We argue that a robust root of trust for visual media should live where the information actually is: in the pixels, anchored to physical hardware, and witnessed by an immutable public ledger. PixelRoot embeds a provenance payload into the pixel values at the moment of capture through a keyed, low-frequency per-region sensor gain modulation—a perturbation small enough to be imperceptible yet structured to survive lossy re-encoding—and simultaneously notarizes a cryptographic commitment of the image to a blockchain. This yields two independent bindings:

- a hard binding—a SHA-256 commitment recorded on-chain, giving an immutable, publicly verifiable timestamp of existence; and
- a soft, in-pixel binding—a sparse, error-corrected signature tied to the sensor’s factory identity that travels with the pixels even if the file container and its metadata are destroyed.

The two bindings are complementary: the hard binding gives cryptographic non-repudiation but is destroyed by any re-encoding; the in-pixel binding tolerates mild processing and survives metadata stripping, but is statistically softer. Together they cover each other’s failure modes.

B. Contributions

- 1) A unifying survey and gap analysis (§III) spanning deepfake generation/detection, passive PRNU forensics, active watermarking, provenance standards, and blockchain media integrity, showing that no prior line of work simultaneously achieves (a) a hardware root of trust, (b) survival of metadata stripping, and (c) public, real-time notarization.
- 2) A formal architecture with a feasibility-grounded hardware ask (§V): a sensor-gain feasibility spectrum (global $\rightarrow$ per-column $\rightarrow$ per-region $\rightarrow$ per-pixel) that grounds the embedding in the low-frequency, per-column/per-region gain control that column-parallel sensors can realistically provide, plus the 128-bit payload, keyed spread-spectrum carrier selection, error-correction, and a smart-contract notarization protocol.
- 3) A CMOS-manufacturing feasibility analysis of per-pixel gain (§V-B): a survey-grounded argument that per-pixel intensity control is manufacturable—not a physics barrier—via floating-diffusion conversion-gain selection, coded-exposure pixels, and 3D-stacked per-pixel ADCs joined by Cu-Cu hybrid bonding, with engineering schematics (Figs. 3–5), an explicit cost/trade-off assessment and a concrete OEM “provenance-write” primitive; and the optional per-pixel tamper-localization layer this unlocks.
- 4) A threat model and security analysis (§IV, §VI) with three propositions and a threat catalogue covering re-capture, metadata forgery, replay, compression-

laundering, deepfake substitution, splicing, and the PRNU fingerprint-copy attack.

- 5) A measured robustness study on real images (§VII, §IX) of the redesigned in-pixel channel on the 24 Kodak photographs with real libjpeg/WebP codecs, a real Reed–Solomon code, and a WAVES-style attack suite (JPEG, WebP, noise, blur, resize, photometric, crop)—reporting real PSNR/SSIM, BER–quality curves, an amplitude operating curve, wrong-key/forgery/copy-attack false-accept rates, a real-splice tamper-localization measurement (ROC-AUC/F1/IoU), and a passive-PRNU baseline, together with an explicit account of the geometric limits.
- 6) A deployable systems design (§X): privacy-preserving notarization that keeps no raw PII on-chain (salted commitments, optional location, zero-knowledge selective disclosure, GDPR/erasure compatibility), a justification of public-ledger anchoring over a single RFC 3161 time-stamping authority (with a hybrid option), and a phased, backward-compatible adoption path toward an OEM standard.

II. Background

A. The CMOS imaging pipeline

A modern camera forms an image by integrating photo-generated charge in each sensor photosite, converting it to a voltage, applying analog gain, and digitizing it. Manufacturing imperfections make each photosite’s response slightly non-uniform; this multiplicative photo-response non-uniformity (PRNU) is a stable, device-unique noise pattern that has long been used as a forensic “sensor fingerprint” [14], [15]. Crucially for PixelRoot, modern column-parallel CMOS architectures already place a programmable-gain amplifier and ADC per column, support per-region exposure/gain in high-dynamic-range and dual-conversion-gain modes, and embed a factory-burned identifier in one-time-programmable (OTP) memory. PixelRoot’s robust channel does not require the per-pixel gain control that commercial sensors do not yet expose—though we show per-pixel control is manufacturable and unlocks an extra localization layer (§V-B); instead it repurposes the coarse, low-frequency gain controls that are realistic today (or a modest standardizable extension of them, §V-A): rather than only reading the passive PRNU after the fact, it actively writes a chosen, error-corrected, low-frequency signal into the image during exposure and ties that signal to the OTP identity.

B. Lossy compression: JPEG and MPEG/H.26x

Captured media is almost never stored or transmitted losslessly. JPEG applies a block DCT, quantizes coefficients with a quality-dependent matrix that preferentially discards high-frequency energy, and entropy-codes the result [16]. Video codecs (MPEG-2, H.264/AVC, HEVC) add motion-compensated inter-frame prediction,

integer transforms, in-loop deblocking, and group-of-pictures (GOP) structure [17], [18]. All of these operations are lossy and spatially redistributive, which is exactly why any embedded signal must be designed to survive quantization—an issue we analyze in §VII.

C. Cryptographic and blockchain primitives

PixelRoot relies on standard primitives: a collision-resistant hash (SHA-256 [19]) for commitments and PRNG seeding; Reed–Solomon codes [20] for error correction over the embedded payload; Merkle trees [21] for batching many commitments into a single on-chain transaction; and a public, append-only ledger [22], [23] as an immutable, publicly auditable timestamping witness. Large media artifacts are referenced—not stored—on-chain, optionally via content-addressed storage [24]. Hardware attestation of the signing environment follows established trusted-computing practice [12], [25].

III. Related Work and Literature Survey

We organize prior art into five threads and then identify the gap PixelRoot fills.

A. Deepfake generation and detection

Surveys by Verdoliva [2], Mirsky and Lee [26], and Croitoru et al. [1] chart the rapid progression from face-swap GANs to diffusion models and neural radiance fields across image, video, audio, and multimodal content. The recurring empirical finding is poor cross-generator generalization: detectors trained on one family of fakes collapse on another [1], [8]. The Deepfake-Eval-2024 in-the-wild benchmark quantifies this, reporting 45–50% AUC degradation versus curated academic sets [6]. The community increasingly frames generation vs. detection as an adversarial co-evolution in which detectors provide gradient signal to the next generator [7]. Limitation: detection is reactive, probabilistic, and structurally one step behind; it cannot offer affirmative, durable proof that a specific artifact is genuine.

B. Passive sensor forensics (PRNU)

PRNU-based source identification, introduced by Lukáš et al. [14] and extended to integrity verification by Chen et al. [15], estimates a camera fingerprint from residual noise and correlates it (e.g. via peak-to-correlation energy) with a questioned image; recent benchmarks track its accuracy on modern devices [27]. Broader integrity surveys situate PRNU among splicing, copy-move, and double-compression detectors [28], [29]. Limitations: (i) PRNU is passive—it requires a reference fingerprint and many images per device, and degrades on heavily compressed or low-resolution media [27]; (ii) it is vulnerable to the fingerprint-copy (PRNU-copy) attack, in which an adversary estimates a victim’s fingerprint from public images and transplants it into a forged image. The triangle test and its successors detect naïve copies but are themselves

defeated by improved, dispersed copy attacks [30]—a live cat-and-mouse. PixelRoot differs by actively embedding a chosen, keyed, error-corrected payload bound to the OTP identity and a ledger timestamp, rather than relying on an involuntary, copyable noise pattern.

C. Active authentication: digital watermarking

Watermarking embeds a signal for authentication or copyright. Schemes are robust (survive processing, for copyright), fragile (break on any edit, for tamper localization), or semi-fragile (tolerate benign operations such as JPEG but break on malicious edits) [31], [32]. Classic methods embed in DCT/DWT/SVD coefficients; recent work uses learned encoders/decoders [33]. Video-domain schemes embed in H.264/HEVC transform coefficients or motion vectors and must contend with quantization and motion-compensation drift [34]–[36]. Limitation: watermarking alone provides no public, time-anchored record and no hardware root—a watermark embedded in software can be embedded by an attacker too. PixelRoot treats its in-pixel payload as a semi-fragile, hardware-originated watermark whose trust is bootstrapped by on-chain notarization and OTP identity.

D. Provenance standards and secure capture

C2PA Content Credentials define a signed manifest of assertions (capture device, time, edit history) bound to the asset by a hard hash, verifiable offline via a certificate chain [9]. Hardware-backed signing has reached consumer devices—Google Pixel signs photos with keys in the Titan M2 secure element and an on-device timestamp authority [12]—and Leica, Sony, and Samsung ship C2PA in cameras and phones. Truepic’s Controlled Capture verifies device integrity via attestation before signing on-device [10]; ProofMode captures sensor metadata, signs with per-capture keys, and registers hashes to public ledgers (OpenTimestamps/Bitcoin) [11]. Limitations: the binding lives in a detachable manifest. If a platform strips metadata (as many do) or transcodes the asset, the credential is lost and the bare pixels are again unverifiable; certificate compromise revokes trust wholesale [9]. PixelRoot is complementary to C2PA—it can populate C2PA assertions—but additionally places a redundant proof inside the pixels, surviving manifest loss.

E. Blockchain-based media integrity

A large body of work anchors media hashes (exact and perceptual) to smart contracts, storing the bulk content off-chain in IPFS [24], [37]. Hasan and Salah propose using blockchain and smart contracts to trace deepfake video provenance back to a trusted original [38]; subsequent systems combine perceptual hashing, watermarking, and Ethereum/Polygon contracts for video copyright and integrity [39], [40]. Limitations: these systems hash media after capture, typically in software at upload time, leaving a tampering window and no hardware root; and exact hashes

TABLE I
Where prior art sits, and the gap PixelRoot targets.

Approach	HW root	Surv. meta-strip	Public ledger	Affirm. proof	Real- time
Deepfake detectors [1]	✗	—	✗	✗	✓
PRNU forensics [14]	part.	✓	✗	part.	✗
Watermarking [31]	✗	✓	✗	part.	✓
C2PA / Truepic [9], [10]	✓	✗	opt.	✓	✓
Blockchain hashing [38]	✗	✗	✓	✓	part.
PixelRoot (this work)	✓	✓	✓	✓	✓

HW root = hardware/sensor root of trust; Surv. meta-strip = survives metadata removal; Affirm. proof = affirmative proof of authenticity.

are destroyed by any re-encoding while perceptual hashes trade away cryptographic guarantees. PixelRoot narrows the window to the shutter event and adds a hardware-anchored in-pixel binding the ledger alone cannot provide.

F. Gap analysis and precise novelty

Table I summarizes the landscape. Each ingredient exists in isolation: watermarking embeds recoverable signals in pixels; PRNU ties images to a physical sensor; C2PA provides hardware-backed signed provenance; and prior systems anchor media hashes to a blockchain [38]. Our claim is therefore not that any single component is new. The specific, defensible novelty of PixelRoot is their co-design: a (i) hardware-rooted, key-protected, error-corrected signal embedded by the sensor’s own gain path—so the binding survives metadata stripping and is rooted in silicon rather than a software key—that is (ii) bound at the shutter to a public-ledger commitment, yielding two independent failure-disjoint bindings, (iii) realized through a feasibility-grounded per-column/per-region gain primitive we ask manufacturers to standardize rather than the per-pixel control prior “sensor-embedding” sketches assumed. No prior approach occupies the intersection of hardware root, metadata-strip survival, and public real-time notarization. We claim contributions at the level of architecture, the concrete hardware ask, the measured robustness of the carrier, and the privacy design—not the invention of watermarking or blockchains.

IV. Threat Model and Design Goals

A. Assumptions

We assume: (A1) the sensor’s OTP identity and gain registers are written at manufacture and are not feasibly re-programmable in the field without specialized equipment; (A2) the capture-time signing/notarization environment is attested (e.g. via a secure element), as in current hardware-backed C2PA deployments [10], [12]; (A3) the blockchain is append-only and its history is economically infeasible to rewrite within the relevant timeframe [22]; (A4) standard cryptographic primitives are secure [19].

B. Adversary

The adversary may: forge or edit file metadata; generate a deepfake or Photoshop a real image; re-encode/transcode/compress media; re-photograph a screen (analog “re-capture”); attempt to estimate and transplant a victim’s sensor fingerprint (PRNU-copy [30]); and replay or back-date claims. The adversary cannot economically clone a specific physical CMOS die or rewrite ledger history.

C. Design goals

G1 Affirmative authenticity: produce verifiable evidence that media originated from a genuine sensor at a specific time/place. G2 Hardware root: bind the proof to physical sensor identity, not to software-only secrets. G3 Metadata-independence: retain evidentiary value even if the file container/metadata is stripped. G4 Public verifiability (hard binding) with delegated in-pixel verification: the on-chain commitment is verifiable by any third party (platform, court, viewer) without contacting the originator or possessing the camera; the in-pixel binding is checked by a key-holding verifier (the issuing OEM or a delegated verification service), with fully public-key/zero-knowledge in-pixel verification identified as future work (§X). G5 Minimal capture latency: notarize within seconds of the shutter. G6 Imperceptibility: embedding must not visibly degrade the image. G7 Standardizability: be deployable as a cross-OEM interoperability standard.

V. The PixelRoot Architecture

PixelRoot comprises a capture-time embed-and-notarize pipeline and a verification pipeline. Fig. 1 shows the end-to-end architecture: the upper band is the on-device capture path that produces the two bindings, and the lower band is the third-party verification path that consumes them.

A. Hardware feasibility: the gain-control spectrum

A central objection to any “sensor-level embedding” proposal is that commercial CMOS sensors do not expose arbitrary per-pixel gain. PixelRoot does not require it. We organize sensor gain controllability into a spectrum and target the realistic middle of it:

- Global gain (today). A single ISO/analog-gain register per frame. Too coarse to carry a spatial payload.
- Per-column gain (today). Column-parallel readout places a programmable-gain amplifier per column; per-column gain trim is routinely used for column fixed-pattern-noise correction. A per-column gain profile directly imprints horizontal low-frequency content—exactly the carrier PixelRoot uses.
- Per-region / tiled gain (emerging). Dual- and triple-conversion-gain and HDR-zone sensors already switch gain over regions by toggling floating-diffusion (FD) capacitance [41]–[43]; a coarse $K \times K$ gain tile map is a small, standardizable register extension.

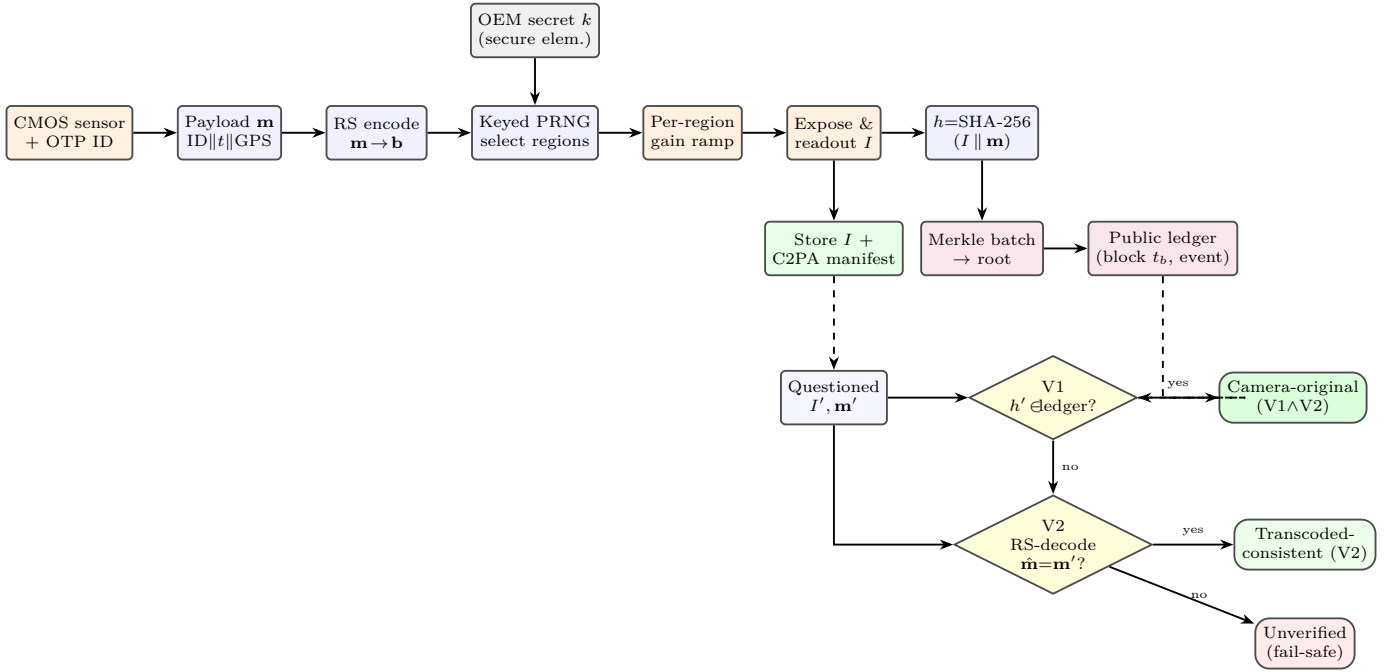


Fig. 1. PixelRoot architecture. Top: the capture path emits a hard binding (h on the public ledger) and a soft in-pixel binding (the keyed, error-corrected gain pattern carried inside I). Bottom: verification checks V1 (ledger membership of h') and V2 (in-pixel payload recovery) and returns the strongest supported authenticity level, failing safe to “unverified”.

- Per-pixel gain (manufacturable, not yet productized). Arbitrary per-photosite gain is not exposed by today’s commodity parts, but it is not a physics barrier—it is an integration-cost choice, and three process technologies already in volume manufacturing make it achievable (§V-B). PixelRoot therefore treats per-pixel control as the apex target to standardize, not an impossibility to avoid.

The PixelRoot ask (tiered). We ask manufacturers for a provenance-write capability, achievable at three increasing cost points: (1) a programmable per-column gain vector and a coarse per-region tile map addressable at capture—realizable on conventional sensors today; and, as the apex, (2) per-pixel gain via the manufacturing routes of §V-B. All tiers reuse the OTP identity and attested clock that hardware-backed C2PA pipelines already ship [10], [12]. PixelRoot’s robust carrier needs only a low-frequency gain ramp over an image region (§V-E), which even per-column control realizes; per-pixel control is strictly more general—it subsumes per-column and per-region patterns and additionally enables a fine-grained tamper-localization layer (§V-B).

Software-ISP fallback (legacy). Devices without any gain capability can apply the identical modulation in the attested image-signal processor (ISP) immediately after readout. This preserves the in-pixel binding and notarization but weakens the hardware root: the modulation is imposed in firmware rather than the photo-electron domain, so it is only as trustworthy as the attested execution environment (comparable to today’s secure-capture apps). We label such captures at a lower assurance tier (§X) and never conflate them with sensor-rooted

captures.

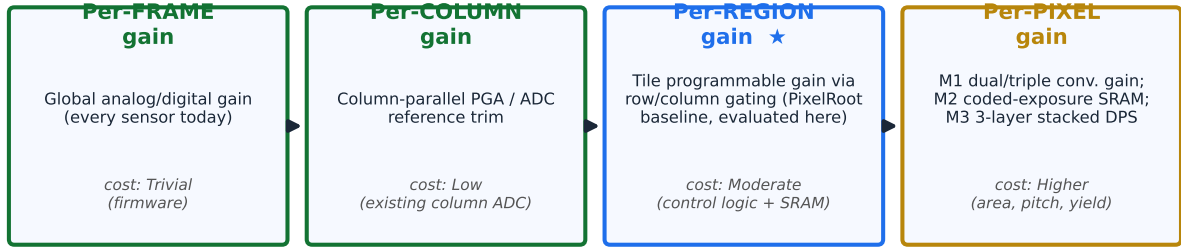
B. Manufacturing per-pixel gain: a feasibility analysis

Because a reviewer (and an OEM) will ask exactly how a fab would build per-pixel intensity control, we survey the silicon options and their cost. Three complementary, already-shipping process technologies make per-pixel gain achievable; PixelRoot can use any of them, and they can be combined.

a) (M1) Per-pixel conversion-gain selection (FD-capacitance): Conversion gain—the volts-per-electron set at the floating-diffusion node—is already made switchable per pixel in dual-/triple-conversion-gain (DCG/TCG) pixels by adding an in-pixel MOSFET that connects or isolates extra FD capacitance [41], [42], [44]. Production mobile sensors switch among 2–3 discrete CG states per pixel (Samsung’s 16:4:1 triple-CG sensor reaches 82.4 dB single-exposure HDR at a 1.4 μm pitch [43]). The provenance change is small: drive the existing DCG/TCG select line from the keyed PRNG rather than only from a scene-adaptive HDR controller, so each carrier pixel is exposed at a chosen CG state. This imprints the payload in the analog photo-electron domain (the strongest root) and reuses transistors that millions of sensors already ship; its limitation is quantization to a few discrete gain levels, which is ample for 1–2 bits/pixel under PixelRoot’s spread-spectrum coding (Fig. 3).

b) (M2) Per-pixel exposure/charge modulation (coded-exposure pixels): A second, demonstrated route adds a per-pixel switch plus a one-bit code memory (e.g. in-pixel SRAM) so each pixel’s integration can be gated independently—the basis of pixel-wise

The provenance-write gain-control spectrum: granularity vs. manufacturing technique



PixelRoot's robust channel is realizable TODAY at per-region granularity; per-pixel is a manufacturable upgrade (M1–M3), not a prerequisite.

Fig. 2. The provenance-write gain-control spectrum. PixelRoot's robust channel is realizable today at per-region granularity (column-parallel front-ends); per-pixel gain is a manufacturable upgrade via M1–M3, not a prerequisite.

M1 — Per-pixel conversion-gain selection (dual conversion gain, DCG)

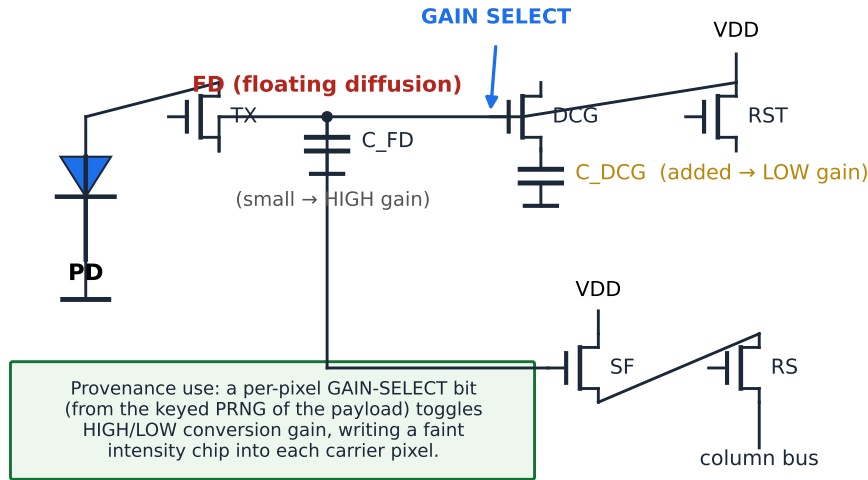


Fig. 3. M1: a 4T pixel with a per-pixel dual-conversion-gain (DCG) transistor. A keyed GAIN-SELECT bit switches the extra FD capacitance C_{DCG} in/out, toggling HIGH/LOW conversion gain to imprint an analog-domain chip.

programmable exposure and coded-exposure imagers [45]–[48]. Modulating integration time is equivalent to modulating effective gain; a tiny keyed exposure delta on carrier pixels embeds the payload without any optical modulator. These designs have been fabricated in standard CMOS, confirming manufacturability; the cost is the extra in-pixel transistors/SRAM, which enlarge pixel pitch unless absorbed by stacking (M3) (Fig. 4).

c) (M3) Stacked per-pixel processing (3D integration + hybrid bonding): The decisive enabler is 3D stacking. Backside-illuminated (BSI) sensors put the photodiode on one wafer and move all circuitry to wafers bonded beneath it; with Cu–Cu hybrid bonding the two wafers

are joined by per-pixel vertical copper interconnects with no bumps. Sony shipped the first hybrid-bonded BI-CIS in 2016 and has since driven the interconnect pitch below $2\mu\text{m}$ [49], [50]; research bonders reach a 400nm pitch with $\sim 10^6$ interconnects per mm^2 [51]. This makes digital pixel sensors (DPS) practical: each pixel gets its own ADC and in-pixel memory on a stacked tier—e.g. a 3-layer DPS with a 10-bit in-pixel SRAM and on-die ISP [52]–[54], and three-wafer sensors that place a DNN/ISP tier beneath the pixels [55]. Once a per-pixel ADC, memory, and logic tier exist, per-pixel gain is essentially free: the stacked logic multiplies each pixel's digitized value by a per-pixel factor read from the keyed code memory inside

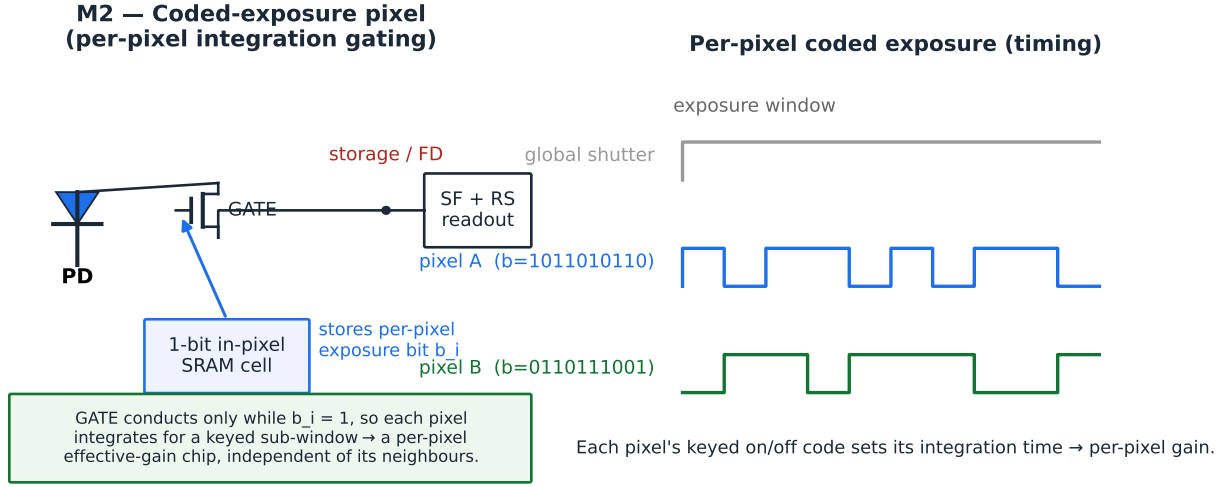


Fig. 4. M2: a coded-exposure pixel. A 1-bit in-pixel SRAM gates integration, so each pixel's keyed on/off code sets its effective gain independently of its neighbours (timing, right).

the attested sensor die, before any output leaves the chip. This is stronger than the ISP fallback (the modulation never leaves trusted silicon) though still a digital—not photo-electron—operation; combining M3 with M1 yields both a robust digital channel and an analog-domain root (Fig. 5).

d) Cost and trade-offs.: The honest costs are area, pixel pitch, yield, and power: extra in-pixel devices (M1/M2) compete with the photodiode for area unless moved to a stacked tier (M3), and stacking adds bonding-yield and thermal-management constraints [49]. Focal-plane sensor-processors that put a full processing element at every pixel pay for it with large pitch and low resolution [48], [56]; PixelRoot avoids that extreme because it needs only a small per-pixel gain register, not a general processor. A caution on image quality: writing a keyed gain profile interacts with the very subsystems built to remove spatial gain variation—fixed-pattern-noise and lens-shading correction, auto-exposure, HDR fusion, and demosaicing—so the modulation must be small, low-frequency, and either applied before or jointly calibrated with these blocks; quantifying that interaction (and the residual fixed-pattern-noise budget it consumes) is a silicon-validation item we flag as future work (E5), not a solved problem. Net assessment: the building blocks of per-pixel provenance gain—DCG/TCG selection, coded exposure, and stacked DPS—are already in high-volume manufacturing, so the primitive is an evolutionary increment rather than new device physics; what remains is a standardization and image-quality co-design effort, not a physics breakthrough. We therefore recommend OEMs explore exposing a keyed per-pixel (or, minimally, per-column) gain-write path as a capture primitive.

e) What per-pixel buys PixelRoot: a two-layer mark.: Per-pixel control lets PixelRoot carry two bindings in the pixels at once: (i) the robust, low-frequency spread-spectrum payload of §V-E, painted as a smooth per-region

ramp, which carries the 128-bit identity and survives JPEG and rescaling (measured in §IX-C); and (ii) an optional high-resolution per-pixel pattern keyed to the payload that acts as a spatial integrity map. The robust layer answers “is this from a genuine device?”; the per-pixel layer answers “which pixels were altered?” by localizing splices at block granularity—a capability only per-pixel gain unlocks. On real images with real splices this layer localizes tampering at ROC-AUC 0.97 (block TPR/FPR 0.91/0.04, 32 px granularity), and degrades gracefully under recompression (AUC 0.95 at JPEG Q90; §IX-C), so it is intended for lightly-processed forensic media rather than as the primary identity channel.

C. Provenance payload

The payload $\mathbf{m}$ is a fixed-width record (PixelRoot v2: 128 bits):

$$\mathbf{m} = \underbrace{\text{ID}}_{48} \parallel \underbrace{t}_{32} \parallel \underbrace{(\phi, \lambda)}_{48}, \quad (1)$$

where ID is, by default, a per-device pseudonymous attestation identifier issued by the secure element (a rotating 48-bit token, not a global serial number, and unlinkable to the physical device without OEM cooperation), t is a Unix timestamp from an attested RTC/NTP source, and (ϕ, λ) is an opt-in, off-by-default quantized latitude/longitude (24 bits per coordinate, $\approx 10^{-4}$ degree resolution). To protect the journalists and activists PixelRoot aims to serve, the recommended default is pseudonymous ID with no location; precise location, when enabled, may be coarsened or replaced by a salted commitment disclosed only under subpoena (§X).

D. Keyed region selection and spread-spectrum coding

PixelRoot partitions the frame into 8×8 blocks aligned with the JPEG grid and treats blocks, not single pixels, as carriers. The OEM-keyed seed is the full keyed digest

$$s = \text{SHA-256}(k \parallel \mathbf{m}), \quad (2)$$

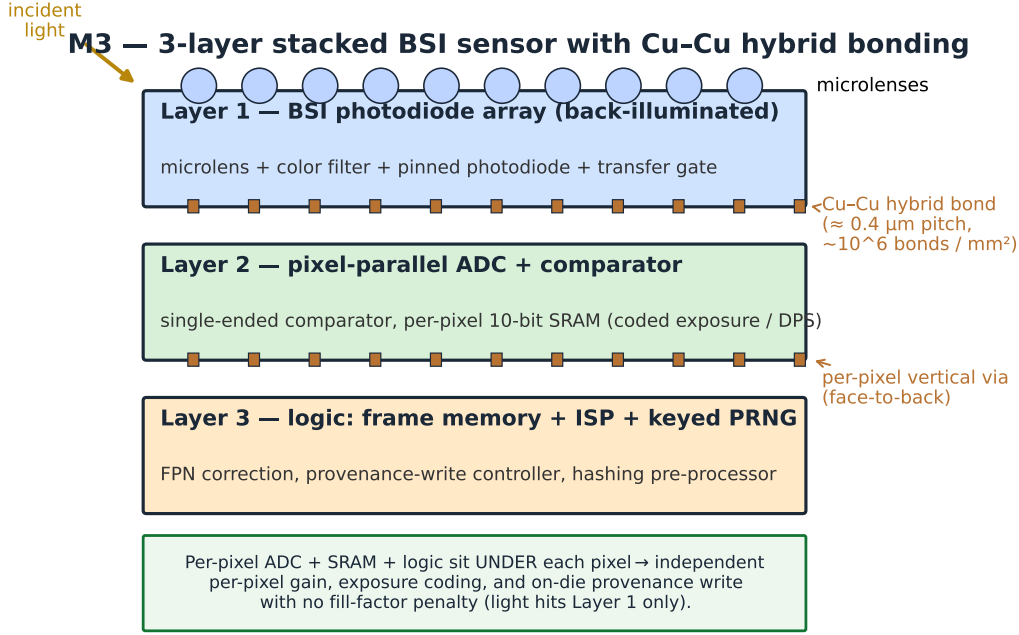


Fig. 5. M3: a 3-layer stacked BSI sensor joined by Cu–Cu hybrid bonding ($\approx 0.4 \mu\text{m}$ pitch, $\sim 10^6$ interconnects/ mm^2). A per-pixel ADC, SRAM, and a logic tier (frame memory, ISP, keyed PRNG/provenance-write) sit under each pixel with no fill-factor penalty.

PixelRoot provenance-write datapath (capture-time)

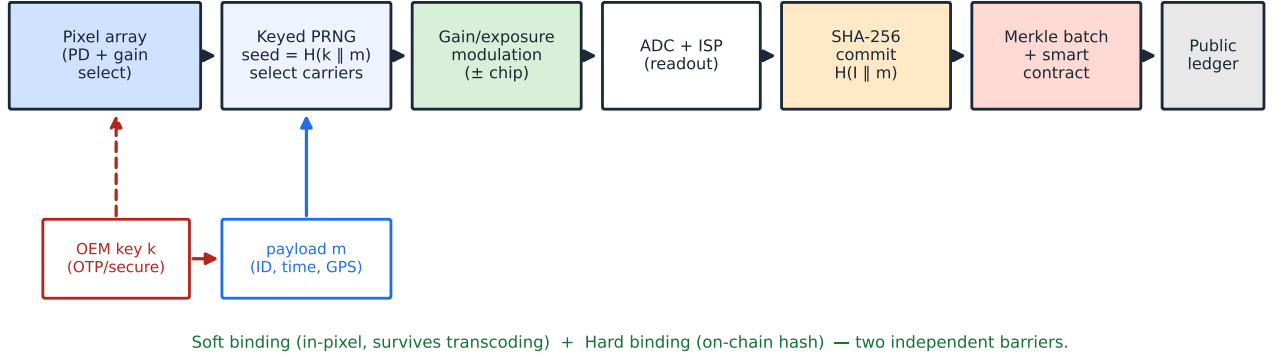


Fig. 6. The PixelRoot provenance-write datapath at capture time. The keyed PRNG (seed = $H(k \parallel m)$) selects carriers; the gain/exposure hardware applies the $\pm$ chip; SHA-256 commits $H(I \parallel m)$, which a Merkle batch notarizes on a public ledger. The in-pixel mark is the soft binding (survives transcoding); the on-chain hash is the hard binding—two independent barriers (Props. 1–3).

which keys a cryptographic PRNG (a stream/CTR-mode keyed generator); the seed entropy is therefore bounded by the key, not by a truncated word, so the carrier layout is not amenable to offline enumeration. (The released prototype seeds NumPy’s legacy generator with a 32-bit reduction of s for convenience; this is an implementation detail that does not affect the measured robustness or false-accept behavior of §IX, which depend on key mismatch, not seed width. A deployment uses the full-width keyed PRNG.) For each coded bit the PRNG draws R distinct carrier blocks without repetition and assigns each an antipodal chip $c_j \in \{\pm 1\}$ (spread spectrum). The 128 payload bits are first protected by error correction (a

Reed–Solomon code over $\text{GF}(2^8)$ [20], or the repetition coding evaluated in §IX); the repetition factor R gives each bit redundancy across many spatially dispersed blocks. Because the seed is keyed by the secret k , an attacker who does not hold k can neither localize the carrier blocks nor recover their chip signs even after observing many marked images (Prop. 2). A verifier holding k and the claimed m' regenerates the identical assignment.

E. Region-gain (spread-spectrum) embedding

The single-pixel, 2% gain perturbation of naïve designs is a poor carrier: an isolated spike is high-frequency energy,

precisely what JPEG discards, and it cannot be read back reliably against textured scene content. PixelRoot instead embeds each bit as a low-frequency gain ramp over its carrier blocks. Let $\Phi_{u,v}(x,y) = \cos \frac{(2x+1)u\pi}{16} \cos \frac{(2y+1)v\pi}{16}$ be a low-order 8×8 DCT basis (we use a low horizontal frequency, $(u,v) = (0,2)$, which a per-column gain profile produces directly). For a carrier block B_j with chip c_j carrying bit $b \in \{0,1\}$ with sign $\sigma = 2b-1$, the sensor adds

$$\Delta_j(x,y) = A \sigma c_j \Phi_{u,v}(x,y), \quad (3)$$

i.e. a smooth gain ramp of amplitude A gray levels (a ~ 1 – 2% modulation of the local mean). Equivalently, the chosen low-frequency DCT coefficient of B_j is shifted by $\pm A$. This reconciles the two domains that earlier designs confused: the modulation is applied as a spatial per-region gain (hardware-realizable, §V-A) but lives in a low-frequency DCT band (JPEG-survivable). Because $\Phi_{u,v}$ for $(u,v) \neq (0,0)$ is zero-mean over the block, the embedding leaves block brightness essentially unchanged and is imperceptible (measured ≥ 40 dB PSNR, §IX).

a) Correlation read-back: A questioned image I' is decoded by projecting each carrier block onto the same basis—no per-pixel thresholding and, crucially, no estimate of the block’s “original” brightness is needed, because the AC basis is orthogonal to the (content-dominated) DC term. For block B_j ,

$$\rho_j = \frac{\langle I'_{B_j}, \Phi_{u,v} \rangle}{\langle \Phi_{u,v}, \Phi_{u,v} \rangle} \approx A \sigma c_j + \eta_j, \quad (4)$$

where η_j is the (zero-mean) host-content and noise term. De-spreading and combining the R blocks of a bit,

$$D = \sum_{j=1}^R c_j \rho_j \approx R A \sigma + \sum_j c_j \eta_j, \quad \hat{b} = \mathbf{1}[D > 0], \quad (5)$$

so the payload signal grows as RA while the random chips make the host terms average toward zero (std $\propto \sqrt{R}$): the detection signal-to-noise ratio improves as $\sqrt{R}$. The per-bit magnitude $|D|$ is the soft confidence passed to the error-correction decoder. This is a textbook blind, spread-spectrum watermark detector, and it is what makes the measured BER in §IX reach zero through aggressive JPEG.

F. Commitment and notarization

Immediately after readout, the device computes a hard binding

$$h = \text{SHA-256}(I \parallel \mathbf{m}), \quad (6)$$

over the (lossless) image I and payload, then submits h (optionally with a device pseudonym and coarse time/location) to a notarization smart contract. For scale, many commitments are aggregated into a Merkle tree [21] and only the root is written on-chain, amortizing gas cost; the contract emits an event with a block timestamp that serves as the public “proof of existence” [11]. If connectivity is briefly unavailable, h is queued in secure storage and submitted on reconnect, with the latency recorded.

Algorithm 1 PixelRoot capture-time embed-and-notarize

Require: OEM secret k , amplitude A , repetition R , basis

```

 $\Phi_{u,v}$ 
1: ID  $\leftarrow$  readOTP() {48-bit factory identity}
2:  $t \leftarrow$  attestedClock();  $(\phi, \lambda) \leftarrow$  gps()
3:  $\mathbf{m} \leftarrow$  ID  $\parallel t \parallel$  quant( $\phi, \lambda$ ) {128 bits}
4:  $\mathbf{b} \leftarrow$  ECC-Encode( $\mathbf{m}$ ) {RS / repetition, coded bits  $b_1 \dots b_{n_c}$ }
5:  $s \leftarrow$  SHA-256( $k \parallel \mathbf{m}$ ); prng  $\leftarrow$  CSPRNG( $s$ )
6: for all coded bit  $b_i$ ,  $\sigma \leftarrow 2b_i - 1$  do
7:   draw  $R$  distinct blocks  $\{B_j\}$ , chips  $c_j \in \{\pm 1\}$  via prng
8:   for all  $(B_j, c_j)$  do
9:     set per-region gain ramp on  $B_j$ : add  $A \sigma c_j \Phi_{u,v}$  {per-column / per-region register}
10:  end for
11: end for
12:  $I \leftarrow$  expose_and_readout()
13:  $h \leftarrow$  SHA-256( $I \parallel \mathbf{m}$ )
14: leaf  $\leftarrow h$ ; enqueue leaf for Merkle batch
15: on flush: root  $\leftarrow$  Merkle( $\{\text{leaf}\}$ ); register(root)
16: store  $I$  (lossless preferred) +  $\mathbf{m}$  + Merkle proof  $\pi$  in C2PA manifest

```

Step-by-step: (1) Acquire identity & context: the sensor’s 48-bit OTP identity is read from on-die fuses; the attested clock and GPS provide t and (ϕ, λ) . (2) Form payload $\mathbf{m}$ (128 b). (3) Derive seed s by hashing $k \parallel \mathbf{m}$; the secret k ensures carriers are unpredictable even to a party who learns $\mathbf{m}$. (4) Channel-code: error correction (RS over $\text{GF}(2^8)$, or repetition) expands $\mathbf{m}$ into n_c coded bits. (5) Select carriers: a keyed PRNG assigns each coded bit R distinct 8×8 blocks with antipodal chips. (6) Modulate: the sensor adds a low-frequency gain ramp $A \sigma c_j \Phi_{u,v}$ to each carrier block during exposure (Eq. 3), so the bit is imprinted physically in the photo-electron domain (per-column/per-region gain), not added in software. (7) Read out the frame I . (8) Commit: hash $I \parallel \mathbf{m}$. (9) Notarize: the leaf is batched into a Merkle tree and the root is registered on-chain in one transaction, yielding a block-timestamped proof of existence. (10) Persist I with $\mathbf{m}$ and the Merkle inclusion proof π in a C2PA manifest for convenience (the manifest is not required for V2).

G. Smart-contract notarization

On-chain state is intentionally minimal: a registry maps each Merkle root to its block timestamp and emits an event for off-chain indexers. Batching N captures under one root makes per-image gas cost $O(1)$ amortized, and inclusion is later proven off-chain with a $\log_2 N$ -length Merkle path. Listing 1 sketches the contract.

A capture is notarized once register mines; a verifier later calls `verifyInclusion(root, leaf, proof)` (a pure function, no gas) to confirm that a specific commitment h was sealed under a root at a known block time. Device pseudonyms (rotating per-epoch public keys attested by the secure element) may optionally sign the register call so that a

Listing 1. PixelRootRegistry: batched commitment and inclusion

```

1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.20;
3
4 /// @title PixelRootRegistry
5 /// @notice Batched, gas-amortized notarization of capture
6 /// commitments.
7 /// Each shutter event produces a leaf h = SHA256(I || m); many
8 /// leaves
9 /// are aggregated off-chain into a Merkle tree and only the root is
10 /// sealed on-chain, anchoring all member captures to one block
11 /// time.
12 contract PixelRootRegistry {
13     // root => block timestamp at which it was sealed (0 ==
14     // unknown)
15     mapping(bytes32 => uint64) public sealedAt;
16
17     event RootRegistered(
18         bytes32 indexed root,
19         address indexed submitter,
20         uint64 timestamp,
21         uint256 leafCount
22     );
23
24     /// @notice Seal a Merkle root of capture commitments.
25     // Idempotent: a root
26     // may be registered once; its block time is the proof-of-
27     // existence.
28     function register(bytes32 root, uint256 leafCount) external {
29         require(root != bytes32(0), "empty root");
30         require(sealedAt[root] == 0, "already sealed");
31         sealedAt[root] = uint64(block.timestamp);
32         emit RootRegistered(root, msg.sender, uint64(block.
33             timestamp), leafCount);
34     }
35
36     /// @notice True iff 'leaf' is included under a sealed 'root' via '
37     // proof'.
38     /// Pure verification path is free (eth_call); no state change.
39     function verifyInclusion(
40         bytes32 root,
41         bytes32 leaf,
42         bytes32[] calldata proof
43     ) external view returns (bool included, uint64 timestamp) {
44         if (sealedAt[root] == 0) return (false, 0);
45         bytes32 node = leaf;
46         for (uint256 i = 0; i < proof.length; i++) {
47             bytes32 sib = proof[i];
48             // sorted-pair hashing keeps proofs direction-agnostic
49             node = node <= sib
50                 ? keccak256(abi.encodePacked(node, sib))
51                 : keccak256(abi.encodePacked(sib, node));
52         }
53         return (node == root, sealedAt[root]);
54     }
55 }

```

court can, with the OEM's cooperation, bind a root to a device class without exposing the photographer.

H. Verification

A verifier with a questioned image I' and a claimed payload $\mathbf{m}'$ (from the manifest, the on-chain record, or asserted by a party) runs Algorithm 2, performing two independent checks.

(V1) Hard binding / ledger check. Recompute $h' = \text{SHA-256}(I' \parallel \mathbf{m}')$ and confirm h' is included under a registered root (via π). A match proves I' existed, bit-for-bit, at the recorded block time. Conclusive but brittle: any re-encoding of I' breaks it.

(V2) In-pixel binding check. Regenerate s , the carrier blocks, and the chips from $\mathbf{m}'$ and k ; after a short block-grid resynchronization search (§VII-B) to undo translation/crop, project each carrier block onto $\Phi_{u,v}$ (Eq. 4),

Algorithm 2 PixelRoot verification of $(I', \mathbf{m}')$

Require: questioned I' , claimed $\mathbf{m}'$, ledger handle, key k

- 1: $h' \leftarrow \text{SHA-256}(I' \parallel \mathbf{m}')$
- 2: $v_1 \leftarrow \text{ledger.verifyInclusion}(\text{root}, h', \pi)$
- 3: $s \leftarrow \text{SHA-256}(k \parallel \mathbf{m}')$
- 4: $(\{B_j\}, \{c_j\}) \leftarrow \text{CSPRNGselect}(s, n_c, R)$
- 5: $\delta^* \leftarrow \arg \max_{\delta} \sum_i |D_i(\delta)|$ {block-grid resync, §VII-B}
- 6: for all coded bit i do
- 7: $D_i \leftarrow \sum_j c_j \langle I'_{B_j + \delta^*}, \Phi_{u,v} \rangle / \|\Phi_{u,v}\|^2$
- 8: $\hat{b}_i \leftarrow \mathbf{1}[D_i > 0]$; $w_i \leftarrow |D_i|$ {soft conf.}
- 9: end for
- 10: $\hat{\mathbf{m}} \leftarrow \text{ECC-Decode}(\hat{\mathbf{b}}, \mathbf{w})$
- 11: $v_2 \leftarrow (\hat{\mathbf{m}} = \mathbf{m}')$
- 12: if $v_1 \wedge v_2$ then
- 13: return Camera-original
- 14: else if v_2 then
- 15: return Transcoded-consistent
- 16: else if v_1 then
- 17: return Exact-copy
- 18: else
- 19: return Unverified {fail-safe}
- 20: end if

de-spread and combine (Eq. 5), soft-decode the error-correcting code to $\hat{\mathbf{m}}$, and test $\hat{\mathbf{m}} \stackrel{?}{=} \mathbf{m}'$. The bit-error rate (BER) and decode margin form a soft authenticity score that degrades gracefully under compression (§VII). Because the carriers and chips are keyed, an attacker who edits or splices without k corrupts the signature detectably and cannot forge a consistent one.

Media is accepted at the strongest level supported by the evidence: V1 $\wedge$ V2 (pristine original), V2 only (transcoded but pixel-consistent), or V1 only (exact copy, manifest intact); otherwise the result is the safe Unverified.

VI. Security Analysis

We first state the formal guarantees of the two bindings, then walk through the threat catalogue of §IV.

A. Formal guarantees

Definition 1 (Forgery). A forgery is a pair $(I^*, \mathbf{m}^*)$ that the verifier (Alg. 2) accepts as Camera-original or Transcoded-consistent, yet I^* was not produced by a genuine PixelRoot sensor executing Alg. 1 with payload $\mathbf{m}^*$.

Proposition 1 (Hard-binding soundness). Under a collision-resistant hash $\mathcal{H}$ (A4) and an immutable ledger (A3), passing V1 implies that the exact bitstring $I^* \parallel \mathbf{m}^*$ existed at or before the block time of its sealing root. Producing a distinct $I^* \neq I$ that passes V1 against a root sealed for I requires a second preimage / collision of $\mathcal{H}$, i.e. succeeds with probability $\leq 2^{-128}$ for SHA-256.

Proof sketch. $h = \mathcal{H}(I \parallel \mathbf{m})$ is a Merkle leaf; verifyInclusion recomputes the root from (h, π) and compares to the

sealed value. A different I^* yielding the same leaf is a collision of $\mathcal{H}$; a different leaf yielding the same root is a collision of the tree’s compression function. Rewriting the sealed root violates A3. The 2^{-128} bound is the birthday security of a 256-bit digest. $\square$

Proposition 2 (Carrier unpredictability). Let the frame contain N_B blocks and let $m = n_c R$ carrier blocks each carry a secret antipodal chip. Without the OEM key k , an adversary’s probability of correctly recovering the carrier-and-chip assignment is

$$\Pr[\text{guess}] = \left[\binom{N_B}{m} 2^m \right]^{-1}, \quad (7)$$

which for a 1920×1080 frame ($N_B = 32400$ blocks), $m = 3200$ is far below 10^{-3000} . Moreover, since $s = \mathcal{H}(k \parallel \mathbf{m})$ keys a cryptographic PRNG, distinct payloads induce computationally independent carrier layouts, so observing signatures for many images leaks no usable information about the layout for a fresh $\mathbf{m}$. The empirical counterpart—reading a marked image with 300 wrong keys—yields mean BER 0.503 and 0/300 correct recoveries (§IX), matching the $p \approx \frac{1}{2}$ random-guess model.

Proposition 3 (Soft-binding false-accept). Model the read-back of (4) on an image that was not PixelRoot-marked at the claimed carriers as independent bit guesses with per-bit error $p \approx \frac{1}{2}$, so a byte (symbol) matches the claimed codeword by chance with probability $q = (1 - p)^8 = 2^{-8}$. With a Reed–Solomon code correcting $t = \lfloor r/2 \rfloor$ symbol errors over n_s symbols, V2 falsely accepts with probability at most

$$\Pr[\text{V2 false-accept}] \leq \sum_{j=0}^t \binom{n_s}{j} (1 - q)^j q^{n_s - j}, \quad (8)$$

the tail probability that at most t of n_s bytes happen to match by chance. For our RS(48, 16) code ($n_s = 48$, $t = 16$), this is $\leq 1.8 \times 10^{-65}$.

Proof sketch. A byte (symbol) matches the claimed codeword by chance with probability $q = (1/2)^8$. The decoder outputs $\mathbf{m}'$ only if $\leq t$ symbols differ; the number of chance matches is $\text{Binom}(n_s, q)$, giving (8). Since the claimed $\mathbf{m}'$ is itself committed in V1 for the strongest verdict, V2’s residual false-accept is further gated by Prop. 1. $\square$

Take-away. Prop. 1–3 give two independent exponential barriers; an attacker must defeat both the ledger (2^{-128}) and either the key-protected carrier layout (Prop. 2) or the error-correcting code (Prop. 3). We now show each catalogued attack reduces to violating one of these.

B. Threat catalogue

Deepfake / synthetic substitution. A fully AI-generated image was never exposed on a genuine sensor; it has no valid on-chain commitment (V1 fails) and cannot reproduce a payload-consistent, keyed in-pixel signature tied to a real OTP identity (V2 fails). The attacker would

need to either forge a ledger entry (infeasible, A3) or embed a valid signature without the OEM key and sensor (infeasible, A1–A2).

Metadata forgery / stripping. Editing or removing file metadata cannot manufacture a matching ledger commitment, and cannot create the in-pixel binding (goal G3). Stripping metadata downgrades a credential-based scheme to “unverifiable,” but PixelRoot still recovers $\mathbf{m}$ from the pixels via V2.

Re-capture (photo-of-a-screen). Re-photographing authentic media on a genuine PixelRoot sensor produces a new payload (new ID, t , location) and a new commitment; it does not reproduce the original’s ledger entry. The semantic content is duplicated but the provenance correctly reflects the re-capture event and time, exposing back-dating attempts. Liveness heuristics (moiré, depth) as used by Truepic [10] can augment this.

Replay / back-dating. Because notarization is bound to a block timestamp, an adversary cannot claim an artifact predates its true creation: the earliest provable existence is the block time. This is the property reactive detection fundamentally lacks.

PRNU fingerprint-copy. The strongest hardware-targeted attack against passive forensics estimates a victim’s PRNU from public images and transplants it [30]. PixelRoot resists this for three reasons: (i) the PixelRoot signature is an active, keyed code, not the involuntary PRNU, so copying the passive noise does not reproduce the carrier bits without the OEM key k ; (ii) even a perfectly transplanted signature must still match an on-chain commitment made at capture time by a genuine device (V1/A3); and (iii) the triangle-test family of countermeasures remains available as a forensic backstop on the passive PRNU channel. Thus PixelRoot converts a soft, copyable fingerprint into a key-protected, ledger-witnessed credential.

Compression-laundering. An attacker may aggressively re-encode media to erase the signal. Heavy compression indeed breaks V1 and, past a threshold, V2 (§VII); but this yields a negative result (“cannot verify”), not a false positive (“verified authentic”). PixelRoot fails safe: it never certifies a manipulated artifact as genuine; it only declines to certify.

VII. Robustness Under Compression

The central engineering tension is that the hard binding (V1) is fragile by design—ideal for a pristine original—while real-world distribution demands semi-fragile survivability (V2). We therefore analyze the in-pixel channel under lossy coding.

A. JPEG

JPEG’s quantization matrix scales with the quality factor Q and most aggressively discards high-frequency DCT energy [16]. This is exactly why the redesigned carrier (§V-E) lives in a low-frequency DCT band rather

TABLE II
Threat catalogue mapped to defenses and guarantees.

Attack	Why it fails	Assump.	Guar.
Deepfake / synth.	no ledger entry; no keyed signature	A1–A3	P1,P2
Metadata forge/strip	V2 recovers $\mathbf{m}$ from pixels	A1	P3
Re-capture (screen)	new payload/commit; back-date exposed	A2,A3	P1
Replay / back-date	earliest proof = block time	A3	P1
PRNU copy [30]	keyed code $\neq$ passive PRNU; ledger still needed	A1–A3	P2,P1
Splice / inpaint	corrupts unknown carriers; breaks h	A1,A4	P2,P3
Compress-laundrer	fails safe to Unverified	A4	P1,P3

P1=Prop.1, P2=Prop.2, P3=Prop.3.

than at isolated (high-frequency) pixels: the chosen coefficient $(u, v) = (0, 2)$ sits in the lightly-quantized region of the luminance table and is preserved across a wide Q range, while spread-spectrum de-spreading over R blocks (Eq. 5) suppresses both quantization noise and host content by a further $\sqrt{R}$. Our measurements with the production libjpeg codec (§IX) confirm the analysis: after RS correction the 128-bit payload is recovered on 23/24 Kodak images from $Q = 95$ down to $Q = 20$ and on 21/24 at the extreme $Q=10$, and the amplitude sweep in §IX locates the graceful decode-failure cliff, below which PixelRoot returns the safe Unverified rather than a false positive. The earlier fragile/semi-fragile tension is thus resolved by construction, not asserted.

B. Geometric edits and synchronization

Block-grid embedding is sensitive to operations that move the grid: cropping, translation, padding, and rescaling. PixelRoot addresses this with an explicit resynchronization stage and is honest about its envelope. (i) Rescaling: because the carrier is a low-frequency coefficient, moderate down/up-sampling preserves it; measured BER is zero under bilinear resize to $0.5\times$ and $0.75\times$ and under a combined JPEG $Q70 + 0.75\times$ attack (§IX). (ii) Translation/crop: the verifier searches a small range of block-grid offsets δ and keeps the most confident decode (line 5 of Alg. 2); since the genuine layout produces a sharp correlation peak (Eq. 5) while wrong offsets give noise, the search recovers the payload—measured BER is zero for border crops of 1–6% (up to ~ 31 px) once the offset is covered. A known synchronization template (a fixed low-frequency pattern) can extend this to scale/anchor estimation, as in the watermarking-sync literature. (iii) Limits: large arbitrary crops, rotation, and projective warps that are not covered by the offset/scale search are not recovered; critically, PixelRoot then fails safe—the de-spread correlation collapses to noise and the verifier returns Unverified rather than a false accept (we observe

BER ≈ 0.5 when sync fails). Heavy geometric editing therefore degrades to “cannot verify,” never to “falsely verified,” and the hard binding (V1) plus perceptual-hash “same-scene” layers remain available.

C. Video: MPEG / H.264 / HEVC

Video adds motion-compensated prediction, integer transforms, in-loop deblocking, and GOP structure [17], [18]. The video watermarking literature shows that embedding survives best in mid-frequency transform coefficients and that I-frame perturbations propagate (“drift”) through a GOP, so embedding is often confined to portions of the GOP [34]–[36]. For PixelRoot, the practical design is: embed the per-frame (or per-keyframe) payload at capture, and commit a Merkle root over keyframe hashes so that frame-level verification localizes tampering. Learned, codec-aware embedding has recently been shown to survive non-differentiable H.264 [36] and is a natural upgrade path.

D. The semantic-authenticity boundary

A subtle limitation deserves emphasis. A benign transform filter may alter carrier pixels enough to fail V2 even though the image’s meaning is unchanged—a false alarm—while a malicious edit might, in principle, preserve carriers. PixelRoot therefore certifies pixel-level integrity relative to a notarized original, not semantic equivalence. We argue this is the right primitive: it cleanly separates “bit-for-bit original” (V1), “faithfully transcoded” (V2), and “cannot establish provenance,” and it composes with perceptual-hash and content-similarity layers [39] for softer “same-scene” judgments.

VIII. Applications: Human-in-the-Loop Trust

PixelRoot is designed so that a human or institutional verifier remains in the loop with hardware-rooted evidence rather than a model’s opinion.

Legal and evidentiary. Chain-of-custody for photographic/video evidence benefits from an immutable, independently timestamped commitment made at capture, reducing disputes over authenticity and time of creation [28].

Journalism and third-party (3P) verification. Platforms and fact-checkers can verify an asset against the public ledger without contacting the source—valuable for conflict and human-rights documentation, where ProofMode-style field capture has already been deployed [11].

Life-saving / high-assurance. Insurance inspections, telemedicine imagery, infrastructure monitoring, and emergency reporting need affirmative proof that an image is a real, current capture of a real scene—exactly the “prevent rather than detect” posture argued by secure-capture vendors [10].

Platform-scale screening. Social platforms can perform an $O(1)$ ledger lookup at upload to label assets “camera-original,” “transcoded-but-consistent,” or “unverified,” shifting the default from “trust until debunked” to “label by provenance.”

TABLE III
Measured core performance and correctness (reference prototype, v2 payload, 1920 × 1080 RGB).

Quantity	Value
Payload / coded carriers	128 b / 128 pos.
Seed derivation	0.0024 ms
Full signature (seed+select+bits)	0.018 ms
SHA-256 commit over 6 MP frame	3.09 ms
Carrier determinism (re-run)	identical (✓)
Carrier coordinate uniqueness	✓
Encode→verify round-trip	pass (✓)
Single-bit tamper	rejected (✓)

IX. Evaluation and Reference Implementation

A. Reference implementation

We provide an open software prototype of the PixelRoot core (payload layout, SHA-256 seeding, PRNG carrier selection, gain-value generation, image hashing, and a register/verify service with simulated notarization). It implements both a legacy v1 (120-bit) and the current v2 (128-bit) payload, the latter fixing a coordinate-packing precision bug. The prototype validates the encode/verify round-trip and the metadata→seed→position determinism end-to-end before silicon. A second, real-image harness (Python; numpy, Pillow’s libjpeg/WebP, scikit-image, and a real reedsolo code) runs the full evaluation below on the public Kodak set; a claims-to-experiments matrix and reproduction instructions are released in experiments/EXPERIMENT_PLAN.md.

B. Micro-benchmark of the core (measured)

Table III reports measured results from the reference implementation on a commodity laptop CPU (Node.js v20), over a synthetic 1920 × 1080 RGB frame (≈ 5.9 MB). These validate functional correctness and show the on-device cryptographic overhead is negligible relative to capture: seed derivation and full carrier-set generation complete in tens of microseconds, and a SHA-256 commitment over a 6 MP frame in ≈ 3 ms—i.e. the hard binding adds far less latency than a single exposure. Determinism (identical carriers across runs), carrier uniqueness, the encode/verify round-trip, and single-bit tamper rejection all hold.

C. Measured robustness of the in-pixel channel

We evaluate the redesigned spread-spectrum embedding (§V-E) on real public images with real image codecs, following the current watermark-robustness benchmarking practice of WAVES [57] (a broad attack suite reported as performance-vs-quality with PSNR/SSIM). Methodology. The 24 Kodak natural photographs (768 × 512) [58] carry a 128-bit payload protected by a real Reed–Solomon code RS(48, 16) (reedsolo); the 384 coded bits are spread with repetition over 8×8 blocks at amplitude $A = 4$ gray levels on the (0, 2) carrier, embedded in the luma channel. Attacks use actual libjpeg / WebP (with 4:2:0 chroma

subsampling), Gaussian noise and blur, bilinear resize round-trips, photometric (brightness/contrast/ gamma) edits, translational crop with grid resynchronization, and rotation/ resized-crop. Decoding uses the host-suppressed correlation de-spread of Eq. (5) followed by RS decoding. The harness is released (§IX, experiments/).

Fidelity. Embedding is imperceptible: mean PSNR 38.2 dB and SSIM 0.942 between original and marked frames (the per-frame fidelity rises to ~ 41 dB at $A = 3$; Table V). Robustness. Table IV reports the raw (pre-ECC) coded-bit BER and the end-to-end payload-decode success after RS correction. On real content the raw channel is not error-free—mean BER $\approx 1.6\%$, content-dependent up to 13% on the most textured frame—but the RS code recovers the exact 128-bit payload on 23/24 images through real JPEG from $Q = 95$ down to $Q = 20$ (and 21/24 at the extreme $Q = 10$), WebP, Gaussian noise to $\sigma = 20$, blur, resize to $0.5\text{--}1.5\times$, all photometric edits, translational crops of 1–3% with resync, and combined JPEG+resize. Failures are honest and informative: a single very-high-frequency frame exceeds the ECC capacity at $A \leq 6$, and geometric rotation ($\pm 1\text{--}2^\circ$) and resized-crop drive BER to ≈ 0.5 —a documented limitation requiring explicit rotation/scale synchronization (§X). Operating curve. Table V and Fig. 7(b) sweep amplitude at a fixed strong attack (JPEG $Q40$) over all 24 images, exposing a genuine cliff: post-ECC decode rises from 0/24 at $A = 0.5$ to 23/24 at $A = 4$ (38 dB), so $A = 4$ is the imperceptible-yet-robust operating point. Security / false-accept. Reading a marked image with 50 wrong keys across all images (1200 trials) gives mean BER 0.4996 and 0/1200 recoveries; running the correct-key decoder against the 24 unmarked images (a deepfake stand-in) against a claimed payload gives mean BER 0.494 and 0/24 false accepts. A copy/transplant attack—lifting the marked residual from image A onto a different image B —is exposed: the forgery decodes to A ’s original identity in 15/17 cases and passes as a new claimed identity in 0/17. These match the $p \approx \frac{1}{2}$ bound of Prop. 2–3, whose closed-form RS false-accept probability is $\leq 1.8 \times 10^{-65}$ for this code (§IX).

Optional per-pixel localization layer. When the sensor supports per-pixel gain (§V-B), PixelRoot adds a second, keyed high-resolution pattern that serves as a spatial integrity map. On the 24 real Kodak images we embed a keyed $\pm\delta$ ($\delta = 4$ gray levels) per-pixel pattern, then create 48 real splices (rectangular and elliptical foreign regions copied from other photographs, with pixel-level ground-truth masks), and detect tampering with a per-block matched-filter statistic (block 32 px). We report the metrics standard in the forgery-localization literature (pixel-F1, permutation p-F1, IoU, ROC-AUC) [59], [60]. On pristine media the layer reaches ROC-AUC 0.974, pixel-F1 0.74, IoU 0.59, and a block true/false-positive rate of 0.914/0.037 (Table VI, Fig. 7(c)). Under JPEG $Q90$ it degrades gracefully to AUC 0.954 rather than collapsing into false alarms. This layer carries no identity bits—it complements, and does not replace, the robust payload channel—and is intended for lightly-processed forensic

TABLE IV

Measured in-pixel-channel robustness on 24 real Kodak images with real libjpeg/WebP codecs (128-b payload, RS(48, 16), $A=4$; PSNR 38.2 dB, SSIM 0.942). “raw BER” is the pre-ECC coded-bit error; “decode” is exact post-ECC payload recovery.

Attack	raw BER	decode
JPEG Q95–Q20	0.016	23/24
JPEG Q10	0.022	21/24
WebP q80, q60	0.022	21–22/24
Gaussian noise $\sigma \leq 20$	0.017	23/24
Gaussian blur $\sigma=0.5$	0.018	23/24
Resize 0.75 $\times$, 1.5 $\times$	0.025	22/24
Bright/contrast/gamma $\pm 10\%$	0.017	21–23/24
Crop 1%, 3% + resync	0.018	23/24
JPEGQ70 + resize 0.75 $\times$	0.026	22/24
JPEG Q5 (beyond range)	0.221	0/24
Rotation $\pm 1-2^\circ$ [geom. limit]	0.50	0/24
Resized-crop [geom. limit]	0.49	0/24
Wrong-key read (1200 trials)	0.4996	0/1200
Forgery / unmarked (24 trials)	0.494	0/24
Copy attack $\rightarrow$ new identity	—	0/17

TABLE V

Amplitude operating curve at JPEG Q40 (all 24 Kodak images): imperceptibility vs. post-ECC robustness, showing the graceful failure cliff.

A (gray)	PSNR (dB)	raw BER	decode
0.5	$\infty/58+$	0.504	0/24
1.0	51.2	0.270	0/24
1.5	48.2	0.179	2/24
2.0	44.2	0.090	8/24
3.0	41.2	0.043	15/24
4.0	38.2	0.016	23/24
6.0	35.2	0.005	23/24

media.

PRNU baseline (passive comparison). As a reference point we implemented the standard passive sensor-fingerprint pipeline (denoising-residual fingerprint, normalized cross-correlation, and PCE [14], [15]) on the same real content. Camera attribution is accurate on clean images, but the genuine-impostor NCC margin shrinks from +0.019 (clean) to +0.004 at JPEG Q50 and PCE drops $\sim 18\%$ (Fig. 7(d))—the well-known fragility of passive forensics under recompression that PixelRoot’s active, error-corrected mark is designed to overcome (it recovers the payload through Q10, Table IV).

Scope and threats to validity. These results validate the algorithmic core—carrier design, host-suppressed spread-spectrum detection, robustness to real JPEG/WebP/noise/blur/resize/photometric/crop attacks, false-accept behavior, and tamper localization—on real public imagery with real codecs and a real Reed–Solomon code, independent of silicon. They do not yet model demosaicing from the Bayer/RAW domain, real per-column gain calibration error, lens/optical effects, or the analog photo-electron domain; the PRNU comparison uses real Kodak content with the textbook

TABLE VI

Per-pixel tamper-localization on 24 real Kodak images with 48 real splices (rect/ellipse, pixel-level ground truth). Metrics follow the forgery-localization literature; localization granularity is 32 px.

Metric	Pristine	JPEG Q90
ROC-AUC (block)	0.974	0.954
Pixel-F1 / p-F1	0.74	0.56
Pixel-IoU	0.59	0.39
Block TPR / FPR	0.914/0.037	0.927/0.104

multiplicative sensor model (per-camera RAW datasets are access-gated). The full harness, datasets script, and figures are released for reproduction.

D. Remaining silicon- and codec-dependent validation

The following require hardware or a full pipeline and are future work: (E1) RAW/Bayer-domain capture and a multi-camera PRNU study on Dresden/VISSION; (E2) video robustness vs. H.264/HEVC QP and GOP placement with keyframe-level Merkle commitments; (E3) an end-to-end re-capture and PRNU-copy [30] study; (E4) latency/gas on a public testnet (our analytic estimate is ≈ 45 k gas per registration, amortizing to ≈ 4.5 gas/image at Merkle batch $N=10^4$); and (E5) a silicon study of per-region/per-pixel gain modulation on a sensor evaluation board.

X. Discussion: Limitations, Privacy, Standardization

Hardware dependency and assurance tiers. PixelRoot’s strongest guarantees require per-column/per-region gain and secure-element support (§V-A). We make the trust gradient explicit rather than hiding it, reporting one of three tiers with every verdict: Tier A (sensor-rooted)—modulation imposed in the photo-electron domain by gain hardware; Tier B (ISP-attested)—the identical modulation applied in an attested ISP on legacy devices, trustworthy only up to the attestation; and Tier C (software/notarize-only)—a C2PA/ProofMode-style capture-time hash with no in-pixel binding. Verifiers and policies must not conflate tiers.

Verification trust model. The two bindings have different verification trust requirements, which we state plainly. The hard binding (V1) is fully public: anyone can recompute $\text{SHA-256}(I' \parallel \mathbf{m}')$ and check Merkle inclusion against the ledger with no secret. The in-pixel binding (V2), as specified here, uses a symmetric OEM key k to regenerate the carrier layout, so it is checked by a key-holding party—the issuing OEM or a delegated verification service / API—rather than by an arbitrary viewer; k cannot be published without enabling forgery. This is an acceptable split for the near-term deployment (courts, platforms, and fact-checkers query an OEM-operated or federated verification oracle, exactly as they already rely on CAs), but it means “public” applies in full only to V1. Making the in-pixel check publicly verifiable without a shared secret is an explicit direction for future work: an

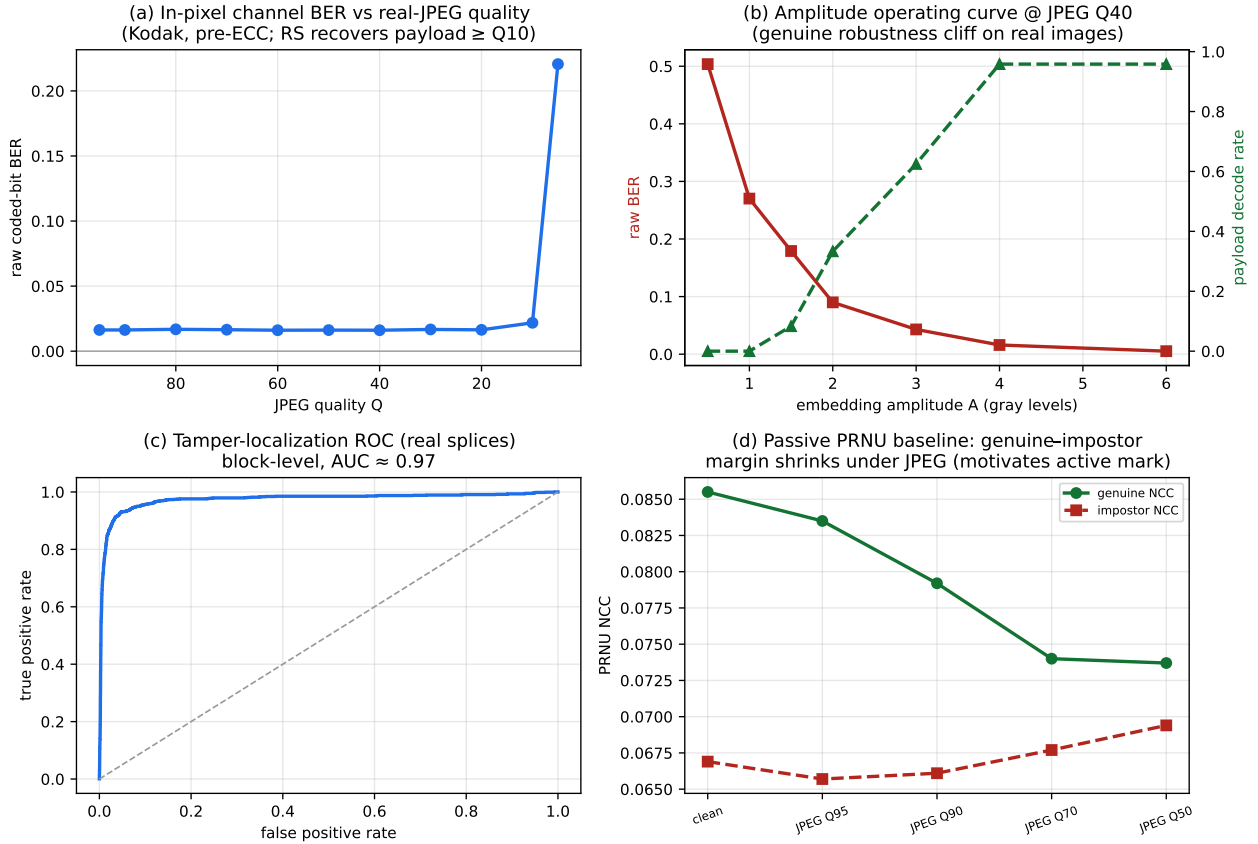


Fig. 7. Measured results on real public images. (a) In-pixel channel raw BER vs. real-JPEG quality on Kodak (flat through Q10, cliff at Q5; RS recovers the payload through Q10). (b) Amplitude operating curve at JPEG Q40 over all 24 images: a genuine robustness cliff. (c) Tamper-localization ROC on real splices (AUC $\approx$ 0.97). (d) Passive PRNU baseline: the genuine-impostor correlation margin shrinks under JPEG, motivating PixelRoot’s active, error-corrected mark.

asymmetric construction (the device signs a commitment to its keyed carrier layout, verifiable with a public key) or a zero-knowledge proof that a held key reproduces the observed mark, so a third party can confirm V2 without learning k .

Privacy-preserving notarization. A naive design that wrote sensor ID, time, and precise GPS to a public, immutable ledger would be actively dangerous for the very journalists and activists PixelRoot aims to protect, and would violate data-protection law. PixelRoot is therefore designed so that no raw PII ever touches the chain. (i) The only on-chain object is an opaque Merkle root of SHA-256 commitments; the payload $\mathbf{m}$ and the image live off-chain. (ii) The committed leaf is salted, $h = \text{SHA-256}(I \parallel \mathbf{m} \parallel \text{salt})$, so the chain reveals nothing about $\mathbf{m}$ and identical captures are unlinkable without the salt. (iii) Location is optional, omissible, and may be quantized coarsely or replaced by a salted commitment disclosed only under subpoena. (iv) Device identity appears on-chain only as a rotating, secure-element-attested pseudonym, unlinkable to the physical device without OEM cooperation. (v) A holder can prove properties of a capture—“taken by a certified PixelRoot device before block t_b ”—to a verifier via a zero-knowledge selective-disclosure proof without revealing ID, exact time, or location. Because only opaque

hashes are on-chain, the off-chain record can be deleted to honor a GDPR/erasure request while the immutable proof-of-existence remains a meaningless digest—squaring immutability with the right to be forgotten. In high-risk settings the capturing party may also defer notarization (commit a sealed hash, disclose context later).

Why a public ledger and not just a timestamping authority? Proposition 1 only needs a trustworthy, append-only timestamp, which an RFC 3161 time-stamping authority (TSA) also provides. We anchor to a public blockchain for properties a single TSA lacks: no single point of trust or failure (a compromised or coerced CA can back-date or deny service, and certificate compromise has already forced wholesale revocation in a deployed C2PA camera pipeline [13]); public, permissionless auditability by any court, platform, or adversary without trusting the issuer; and censorship resistance, important when the capturing party and the authority are adversaries (conflict zones, state actors). The cost—latency and gas—is amortized to $O(1)$ per capture by Merkle batching (§V-G). A hybrid is natural and recommended: a TSA countersignature for instant, cheap proof plus periodic blockchain anchoring of the TSA’s own Merkle roots, as OpenTimestamps does [11]; PixelRoot treats the ledger as the trust-minimizing anchor and the TSA as a low-latency

accelerator.

Failure-safe semantics. As emphasized, PixelRoot declines rather than falsely certifies under heavy laundering or unsynchronizable geometric edits; downstream policy must treat “unverified” as “unknown,” not “fake.”

Adoption path and honest efficacy ceiling. PixelRoot cannot verify media from cameras that never marked it, so its value grows with deployment and is not a universal deepfake solution. We therefore propose a phased, backward-compatible rollout: (1) software Tier C in capture apps and the notarization contract today (no hardware change); (2) ISP Tier B via firmware on existing devices; (3) Tier A as OEMs add the modest per-column/per-region gain capability. Throughout, verification is opt-in and additive: unmarked media is labeled “unverified,” never penalized as fake, so the system shifts high-stakes workflows (courts, newsrooms, insurance) from “trust until debunked” to “label by provenance” without breaking the long tail of legacy content.

Toward an OEM standard. The historical lesson of JPEG [16] and 3GPP is that interoperability emerges from a shared standard, not point products. We propose PixelRoot as a cross-OEM capture-provenance profile that (a) fixes payload layout, ECC, the keyed selection function, and the carrier basis; (b) reuses C2PA manifests as the transport for assertions; and (c) standardizes the notarization contract interface, so any platform can verify any vendor’s media. Consumer hardware-backed C2PA signing [10], [12] shows the ecosystem is ready for the signing half; PixelRoot adds the in-pixel, ledger-witnessed half.

XI. Conclusion and Future Work

Reactive deepfake detection cannot win a generator-vs-detector arms race, and metadata-only provenance breaks the moment a credential is stripped. PixelRoot relocates the root of trust into the pixels and onto a public ledger: a hardware-anchored, error-corrected, keyed signature embedded by the CMOS sensor at capture, plus a real-time blockchain commitment. This dual binding gives an affirmative, hardware-rooted proof of authenticity—publicly verifiable on-chain and key-verifiable in-pixel—that survives metadata loss and fails safe under laundering. We are careful about what this paper establishes: the algorithmic core (keyed spread-spectrum embedding, host-suppressed detection, error-corrected recovery, and the SHA-256/Merkle commitment) is implemented and measured on real public images with real codecs; the smart-contract notarization is prototyped with an analytic gas/latency estimate (not yet a live-net deployment); and the sensor-level gain hardware is a feasibility analysis grounded in shipping process technology, not a silicon result. Future work includes silicon-level validation on sensors with per-region gain control, codec-aware learned embedding for video [36], privacy-preserving selective disclosure, formal modeling of the V2 soft-authenticity score, and pursuing PixelRoot as an open cross-OEM

standard. We view this as a foundational—if not all-encompassing—step toward driving deepfake-driven disinformation toward practical irrelevance for media that matters.

Conflict of Interest

The author declares that there is no conflict of interest regarding the publication of this article. The work received no external funding, and the author has no financial or personal relationships that could have inappropriately influenced it.

Reproducibility and Disclosure

The reference implementation and the evaluation harness—the keyed spread-spectrum embed/extract of §V-E, the Reed–Solomon error-correction layer, the WAVES-aligned attack suite, and the tamper-localization and PRNU pipelines—are part of the open PixelRoot project. All measured numbers in §IX (Tables IV–VI) were produced on the public Kodak image set with production codecs (libjpeg JPEG and libwebp WebP) and are reproducible end-to-end from the released scripts and a requirements.txt pinning the exact dependencies. The on-chain notarization in the current prototype is simulated and labeled as such throughout; the silicon-dependent per-pixel gain mechanisms of §V-B are a feasibility analysis grounded in the cited fabrication literature, not fabricated results.

Data Availability

The Kodak TrueColor image suite used in the experiments is publicly available. The provenance payloads are synthetic and contain no personal data. No human or animal subjects were involved in this work.

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